SE Lab 4: VibeCoding

Dino Run Repair Lab: Lab Report

Name: Vismay Chandra Dev

SRN: PES2UG24CS597

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Assigned repo: github.com/SETAPESU26/36_dinoRun (Pygame Dino Run)

My repo: github.com/VismayCD/36_dinoRun

AI tool used: Claude (chat history: https://claude.ai/share/412758aa-f74e-469b-b329-e4b75c089973)

1. Objective

The goal of this lab was to use an AI assistant as a pair-programming partner to fix one deliberate bug in a Pygame endless-runner game and to implement three optional feature hooks, writing effective prompts and critically reviewing each suggestion rather than accepting it blindly.

2. Summary of Tasks

Task	Problem / requirement	Solution
1. High-score bug	High score never updated or saved, because the comparison in Game.update was backwards.	Changed the condition so the high score updates only when the finished run beats it, then saves it.
2. dino_tint()	Return a colour override based on whether the dino is on the ground.	Returns orange (255, 165, 0) while airborne and None (default green) on the ground.
3. on_obstacle_passed()	Hook called once per obstacle successfully jumped.	Maintains a combo counter of consecutive obstacles cleared; reset on restart and on collision; shown in the HUD.
4. max_jumps()	Return the number of jumps allowed before landing.	Returns 2, enabling a double jump.

3. Before the Changes

I cloned the repository, installed Pygame (pip install pygame) and ran python game.py. The game launches and is playable, but the high score is broken. In the screenshot below I reached a score of 47 and hit an obstacle, yet "Best" still shows 0 on the Game Over screen.



Figure 1: Before. Score 47 at Game Over, but Best remains 0 (bug).

The start screen also showed the dino sunk below the ground line before the first jump, because it was created at y = GROUND_Y instead of resting on top of the ground.



Figure 2: Before. Dino rendered below the ground line on the start screen.

4. Changes Made

Task 1: Fix the high-score bug

Root cause: after a collision, the code checked whether the stored best score was greater than the current run (self.high_score > current). Since the best starts at 0 and the current score is never lower than that, the condition was never true, so the high score never updated and nothing was ever saved to dino_highscore.txt. The comparison sides were swapped.

```
# Before (buggy)
if self.high_score > current:
    self.high_score = current
    save_high_score(self.high_score)

# After (fixed)
if self.high_score < current:   # equivalent to: current > self.high_score
    self.high_score = current
    save_high_score(self.high_score)
```

Task 2: dino_tint(on_ground)

The dino changes colour while it is in the air, which gives clear visual feedback for jumps (and for double jumps).

```
def dino_tint(on_ground):
    if not on_ground:
        return (255, 165, 0)   # orange while airborne
    return None                # default green on the ground
```

Task 3: on_obstacle_passed(obstacle, score)

A module-level combo counter increases each time an obstacle is cleared. It is reset to 0 when the game restarts (Game.reset) and when the dino collides, and it is shown in the score line as "Combo: N".

```
combo = 0

def on_obstacle_passed(obstacle, score):
    global combo
    combo += 1

def reset_combo():
    global combo
    combo = 0

# In Game.reset() and on collision in Game.update():
reset_combo()

# In Game.draw():
f"Score: {self.score // 10}   Best: {self.high_score}   Combo: {combo}"
```

Task 4: max_jumps()

Returning 2 lets the player jump once more in mid-air. The existing jump_count logic in Dino.jump (jump_count < (max_jumps() or 1)) enforces the limit, and Dino.update resets jump_count to 0 on landing, so only the limit had to change.

```
def max_jumps():
    return 2   # double jump
```

Additional small fixes

- Dino is now created at GROUND_Y - 60 so it rests on the ground line from the start screen instead of appearing below it.
- The score text was moved left (WIDTH - 360) so the added "Combo" label fits on screen.

5. After the Changes

After the changes the game shows the combo counter, the dino turns orange in the air, and the high score now updates when a run beats it.

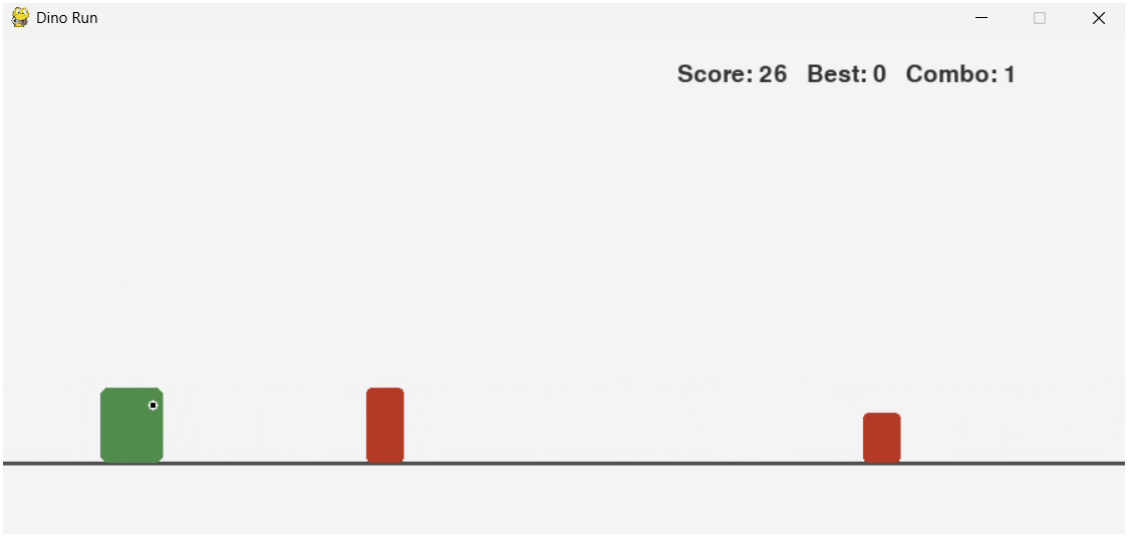


Figure 3: After. Dino on the ground (green) with Combo: 1 after clearing an obstacle.

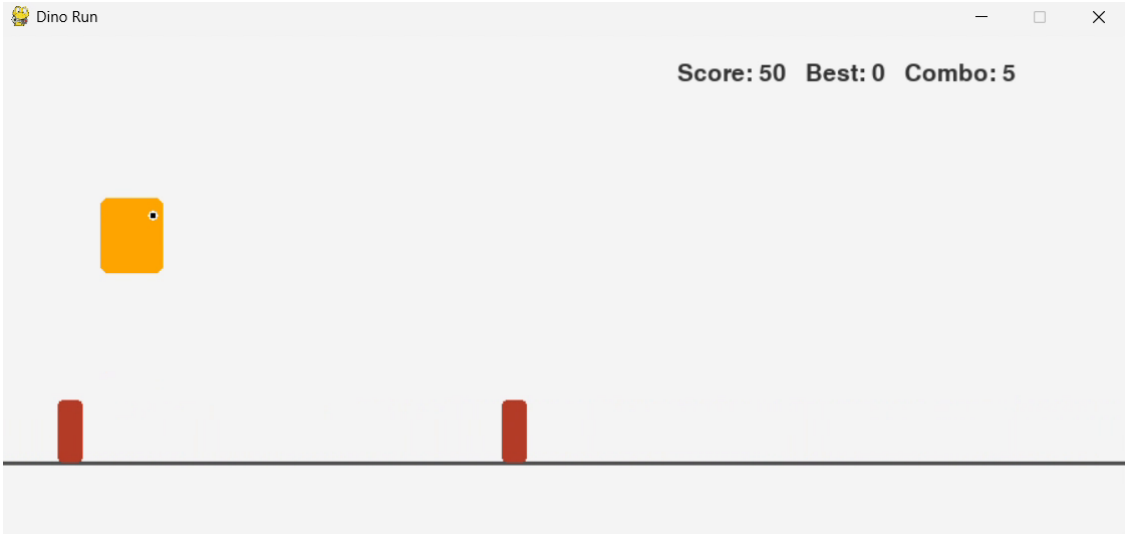


Figure 4: After. Dino is orange while airborne; Combo: 5.



Figure 5: After. Game Over with Score 55 and Best updated to 55 (bug fixed).

Double jump: in the after-video, the dino rises to about 300 px above the ground, almost twice the maximum height of a single jump (about 163 px from JUMP_FORCE = -14 and GRAVITY = 0.6); at the top of the arc it briefly leaves the window. This is only possible with a second jump in mid-air. Figure 6 shows the dino already well above single-jump height. The Best score of 55 from the earlier run is also loaded on startup, which confirms the high score is saved to and read from disk.

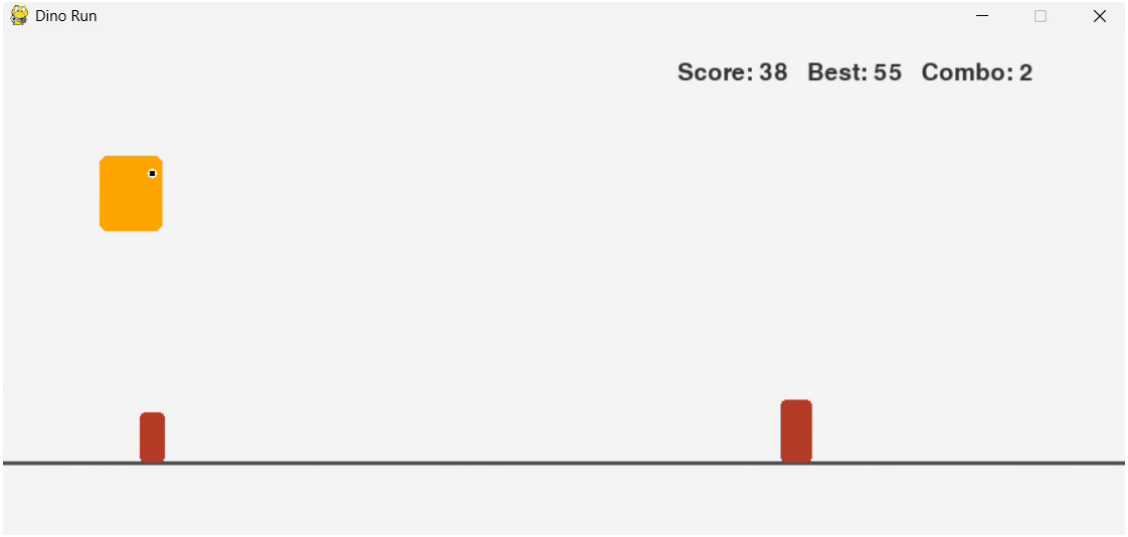


Figure 6: After. Double jump: dino (orange) about 185 px up, above the 163 px single-jump limit; Best: 55 loaded from the previous run.

6. Results

Expected behaviour	Before	After
High score updates when beaten	No (stays 0)	Yes
High score saved to dino_highscore.txt	No	Yes
Colour change while airborne	No	Yes (orange)
Combo counter for cleared obstacles	No	Yes
Double jump allowed	No (1 jump)	Yes (2 jumps)

7. Learnings

- Giving the AI the full source file plus a precise symptom ("high score never updates") led straight to the root cause instead of generic advice.
- Asking for one task per prompt and only the changed lines kept the answers short and easy to review.
- Running the game after every change was essential; the AI output still needs to be tested and read critically.
- Small hook functions (dino_tint, on_obstacle_passed, max_jumps) are a clean way to add features without touching the core game loop.

8. Prompts Used

Four prompts were sent in a single chat, one task per prompt. The full conversation is in the linked chat history.

- **Prompt 1:** Pasted the full game.py and reported that the high score never updates even when the previous best is beaten; asked for the bug in Game.update, why it happens, and the minimal fix.
- **Prompt 2:** Asked to implement dino_tint(on_ground) so the dino turns orange (255, 165, 0) while airborne and returns None on the ground, showing only the changed function.
- **Prompt 3:** Asked to implement on_obstacle_passed as a combo counter of consecutive obstacles cleared, reset in Game.reset() and on collision, displayed as "Combo: N" next to the score in Game.draw, showing only the changed lines and where they go.
- **Prompt 4:** Asked to implement max_jumps() for a double jump, check that jump_count resets on landing, and say whether anything else in Dino.jump or Dino.update needs to change.

9. Deliverables

- Before and after gameplay videos: before_video.mp4, after_video.mp4
- Updated code: game.py (updated version)
- Full chat history: https://claude.ai/chat/caf27463-4973-4871-a216-62b95f676758